

Unbiased Estimators & Confidence Intervals



Point estimates for the mean and variance

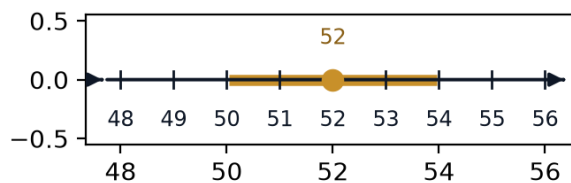
- 1 A sample of 8 measurements has values 12, 15, 11, 14, 13, 16, 12, 15. Find the unbiased point estimate of the population mean.
- 2 Explain why the unbiased estimator of population variance divides the sum of squared deviations by $n - 1$ rather than n , using the formula $s^2 = 1/(n - 1) \sum (x_i - \bar{x})^2$.

Computing the unbiased sample variance

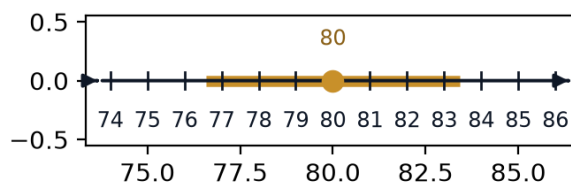
- 3 For the sample 4, 7, 9, 6, 9, find the unbiased estimate of the population variance.

Constructing a confidence interval for the mean

- 4 A sample of $n = 36$ has mean $\bar{x} = 52$ and population standard deviation $\sigma = 6$ (known). Construct a 95% confidence interval for the population mean, using $z^* = 1.96$.

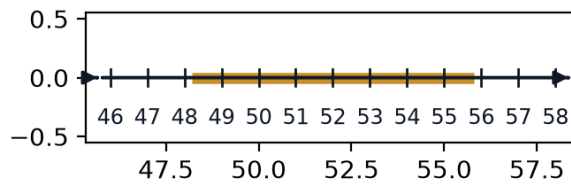


- 5 A sample of $n = 25$ has mean $\bar{x} = 80$ and unbiased sample standard deviation $s = 10$ (population standard deviation unknown, so a t -distribution is used). Construct a 90% confidence interval, using $t^* \approx 1.711$ (with $df = 24$).



Interpreting a confidence interval

- 6 A 95% confidence interval for a population mean is calculated as (48.2, 55.8). State the correct interpretation of this interval, and identify a common misinterpretation to avoid.



Sample size and interval width

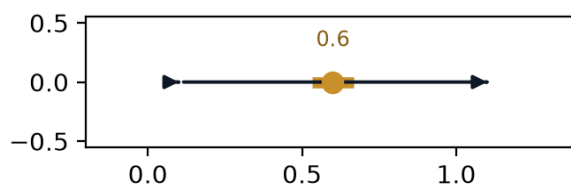
- 7 A researcher wants to halve the margin of error of a confidence interval for the mean, keeping the confidence level and σ fixed. Explain how the required sample size n must change, using the formula for margin of error $E = z^* \sigma / (\sqrt{n})$.

Determining the required sample size

- 8 A researcher wants a 95% confidence interval for a population mean with a margin of error no larger than 2, given a known population standard deviation $\sigma = 9$. Find the minimum required sample size, using $z^* = 1.96$ and rounding up to the nearest whole number.

Confidence intervals for a proportion

- 9 A survey of 200 randomly selected voters finds that 120 support a proposal. Construct a 95% confidence interval for the true population proportion, using $\hat{p} \pm z^* \sqrt{\hat{p}(1 - \hat{p})/n}$ with $z^* = 1.96$.



Comparing pooled vs. unbiased variance estimates

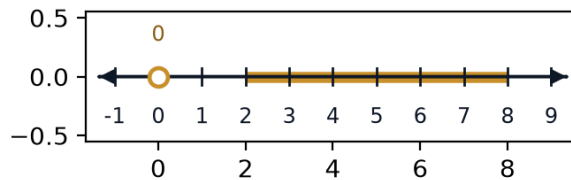
- 10 Two independent samples give unbiased variance estimates $s_1^2 = 16$ (from $n_1 = 10$) and $s_2^2 = 25$ (from $n_2 = 15$). Find the pooled variance estimate, using $s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2) / (n_1 + n_2 - 2)$.

Effect of confidence level on interval width

- 11 Explain why a 99% confidence interval is always wider than a 95% confidence interval for the same sample data, referencing the critical value z^* used in each.

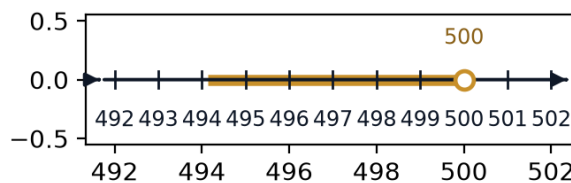
Constructing a confidence interval for a difference of means

- 12 Two independent samples give $\bar{x}_1 = 45$ (with $n_1 = 40$, $\sigma_1 = 8$) and $\bar{x}_2 = 40$ (with $n_2 = 50$, $\sigma_2 = 6$). Construct a 95% confidence interval for $\mu_1 - \mu_2$, using $z^* = 1.96$ and the standard error formula $\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$.



Assessing whether a claimed population mean is plausible

- 13 A machine is supposed to fill bottles with 500ml of liquid. A sample of $n = 30$ bottles has mean $\bar{x} = 497$ ml and sample standard deviation $s = 8$ ml. Construct a 95% confidence interval for the true mean fill volume (using $z^* \approx 1.96$ as an approximation for the t -distribution with $df = 29$), and use it to comment on whether the machine's target of 500ml is plausible.



- 14 Explain why increasing the sample size n , while keeping everything else fixed, makes a confidence interval narrower, referencing the standard error formula $\sigma/(\sqrt{n})$.

Interpreting overlapping confidence intervals

- 15 Two independent 95% confidence intervals for population means are (10, 14) and (13, 17). Explain what can and cannot be concluded about whether the two population means are different, given that the intervals overlap.

Identifying a biased estimator

- 16 A student proposes using the sample maximum as an estimator for the population mean. Explain why this estimator is biased.

Sensitivity of the mean to outliers

- 17 A sample of 5 values is 4, 5, 6, 7, 50, where the last value is a data-entry error that should have been 5. Find the unbiased point estimate of the mean with and without correcting the error, and comment on how sensitive this estimator is to outliers.

One-sided confidence bounds

- 18 Explain how a one-sided (upper) confidence bound differs from a standard two-sided confidence interval, and give its general form using z^* .

Properties of the sample mean as an estimator

- 19 Which of the following statements about the sample mean $\bar{X}$ as an estimator of the population mean μ is correct?
- a. $\bar{X}$ is unbiased, since $E(\bar{X}) = \mu$ for any sample size.
 - b. $\bar{X}$ is biased for small sample sizes but becomes unbiased as $n \rightarrow \infty$.
 - c. $\bar{X}$ is unbiased only when the population is normally distributed.
 - d. $\bar{X}$ is biased because it is calculated from a subset of the population.